1.29 Binomial Expansion

Worked Solutions

1 a)

$\frac{8\times7\times6!}{6!} = 56.$

1 b)

$\frac{50\times49\times48\times47!}{47!\times6} = 50 \times 49 \times 8 = 19600.$

1 c)

$\frac{50\times49\times48\times47!}{47!\times24} = 50 \times 49 \times 2 = 4900.$

1 d)

$\frac{7\times6!}{6!\times1!} = 7.$

1 e)

$\frac{100\times99\times98!}{98!\times2!} = 50 \times 99 = 4950.$

1 f)

$\frac{100\times99!}{99!\times1!} = 100.$

1 g)

$\frac{50\times49!}{49!\times1!} = 50.$

1 h)

$\frac{102!}{102!\times1!} = 1.$

2 a)

$\frac{n\times(n-1)!}{(n-1)!} = n.$

2 b)

$\frac{n\times(n-1)\times(n-2)!}{2\times(n-2)!} = \frac{1}{2}n(n-1).$

2 c)

$\frac{n!}{1!\times(n-1)!} = n.$

2 d)

$\frac{(n+1)!}{n!(n+1-n)!} = \frac{(n+1)\times n!}{n!} = n+1.$

2 e)

$\frac{n!}{n!\times1!} = 1.$

2 f)

$$\frac{(2n-1)!}{2!(2n-1-2)!} = \frac{(2n-1)!}{(2n-3)!2!} = \frac{(2n-1)(2n-2)(2n-3)!}{(2n-3)! \times 2}$$
$$= \frac{(2n-1)(2n-2)}{2} = (2n-1)(n-1)$$

2 g)

$$\frac{(2n-4)!}{(2n-6)!(2n-4-(2n-6))!} = \frac{(2n-4)!}{(2n-6)!2!}$$
$$= \frac{(2n-4)(2n-5)(2n-6)!}{(2n-6)! \times 2}$$
$$= \frac{(2n-4)(2n-5)}{2} = (2n-5)(n-2)$$

3)

$$10 \times \frac{n!}{5!(n-5)!} = 21 \times \frac{n!}{3!(n-3)!}$$
$$60 \times \frac{1}{(n-5)!} = 2520 \times \frac{1}{(n-3) \times (n-4) \times (n-5)!}$$
$$60 = 2520 \times \frac{1}{(n-3)(n-4)}$$
$$(n-3)(n-4) = 42$$
$$n^2 - 7n - 30 = 0$$
$$n = 10 \text{ or } n = -3, \text{ but } n \in \mathbb{N} \text{ so } n = 10$$

4 a)

$$3 \times \frac{n!}{6!(n-6)!} = 11 \times \frac{n!}{4!(n-4)!}$$
$$3 \times \frac{1}{6 \times 5 \times (n-6)!} = 11 \times \frac{1}{(n-4)(n-5)(n-6)!}$$
$$(n-4)(n-5) = 110 \text{ so } n^2 - 9n - 90 = 0$$
$$n = 15 \text{ or } n = -6 \text{ but } n \geq 4 \text{ so } n = 15$$

4 b)

$$28 \times \frac{n!}{4!(n-4)!} = 15 \times \frac{n!}{6!(n-6)!}$$
$$28 \times \frac{1}{4! \times (n-4)(n-5)(n-6)!} = 15 \times \frac{1}{6 \times 5 \times 4!(n-6)!}$$
$$56 = (n-4)(n-5) \text{ so } n^2 - 9n - 36 = 0$$
$$n = 12 \text{ or } n = -3 \text{ but } n \geq 4 \text{ so } n = 12$$

4 c)

$$\begin{aligned}
33 \times \frac{n!}{2!(n-2)!} &= 2 \times \frac{n!}{5!(n-5)!} \\
33 \times \frac{1}{2! \times (n-2)(n-3)(n-4)(n-5)!} &= 2 \times \frac{1}{5 \times 4 \times 3 \times 2!(n-5)!} \\
990 &= (n-2)(n-3)(n-4) \\
n^3 - 9n^2 + 26n - 1014 &= 0 \\
n &= 13
\end{aligned}$$

5 a)

$$\begin{aligned}
(1+x)^6 &= \binom{6}{0}1^6x^0 + \binom{6}{1}1^5x^1 + \binom{6}{2}1^4x^2 + \binom{6}{3}1^3x^3 \\
&\quad + \binom{6}{4}1^2x^4 + \binom{6}{5}1^1x^5 + \binom{6}{6}1^0x^6 \\
&= 1 + 6x + 15x^2 + 20x^3 + 15x^4 + 6x^5 + x^6
\end{aligned}$$

5 b)

$$\begin{aligned}
(2+x)^5 &= \binom{5}{0}2^5x^0 + \binom{5}{1}2^4x^1 + \binom{5}{2}2^3x^2 + \binom{5}{3}2^2x^3 \\
&\quad + \binom{5}{4}2^1x^4 + \binom{5}{5}2^0x^5 \\
&= 32 + 80x + 80x^2 + 40x^3 + 10x^4 + x^5
\end{aligned}$$

5 c)

$$\begin{aligned}
(x+3)^6 &= \binom{6}{0} \times x^6 \times 3^0 \\
&\quad + \binom{6}{1} \times x^5 \times 3^1 + \binom{6}{2} \times x^4 \times 3^2 \\
&\quad + \binom{6}{3} \times x^3 \times 3^3 + \binom{6}{4} \times x^2 \times 3^4 \\
&\quad + \binom{6}{5} \times x^1 \times 3^5 + \binom{6}{6} \times x^0 \times 3^6 \\
&= 729 + 1458x + 1215x^2 + 540x^3 + 135x^4 + 18x^5 + x^6
\end{aligned}$$

5 d)

$$\begin{aligned}(x-1)^4 &= \binom{4}{0} \times x^4 \times (-1)^0 \\ &\quad + \binom{4}{1} \times x^3 \times (-1)^1 + \binom{4}{2} \times x^2 \times (-1)^2 \\ &\quad + \binom{4}{3} \times x^1 \times (-1)^3 + \binom{4}{4} \times x^0 \times (-1)^4 \\ &= 1 - 4x + 6x^2 - 4x^3 + x^4\end{aligned}$$

6 a)

$$\begin{aligned}(x-3)^5 &= \binom{5}{0} \times x^5 \times (-3)^0 \\ &\quad + \binom{5}{1} \times x^4 \times (-3)^1 + \binom{5}{2} \times x^3 \times (-3)^2 \\ &\quad + \binom{5}{3} \times x^2 \times (-3)^3 + \binom{5}{4} \times x^1 \times (-3)^4 \\ &\quad + \binom{5}{5} \times x^0 \times (-3)^5 \\ &= x^5 - 15x^4 + 90x^3 - 270x^2 + 405x - 243\end{aligned}$$

6 b)

$$\begin{aligned}(5-x)^6 &= \binom{6}{0} \times 5^6 \times (-x)^0 \\ &\quad + \binom{6}{1} \times 5^5 \times (-x)^1 + \binom{6}{2} \times 5^4 \times (-x)^2 \\ &\quad + \binom{6}{3} \times 5^3 \times (-x)^3 + \binom{6}{4} \times 5^2 \times (-x)^4 \\ &\quad + \binom{6}{5} \times 5^1 \times (-x)^5 + \binom{6}{6} \times 5^0 \times (-x)^6 \\ &= x^6 - 30x^5 + 375x^4 - 2500x^3 + 9375x^2 - 18750x + 15625\end{aligned}$$

6 c)

$$\begin{aligned}(2x+1)^4 &= \binom{4}{0} \times (2x)^4 \times 1^0 \\ &\quad + \binom{4}{1} \times (2x)^3 \times 1^1 + \binom{4}{2} \times (2x)^2 \times 1^2\end{aligned}$$

$$
\begin{aligned}
&+ \binom{4}{3} \times (2x)^1 \times 1^3 + \binom{4}{4} \times (2x)^0 \times 1^4 \\
&= 16x^4 + 32x^3 + 24x^2 + 8x + 1
\end{aligned}
$$

6 d)

$$
\begin{aligned}
(3x-3)^5 = &\binom{5}{0} \times (3x)^5 \times (-3)^0 \\
&+ \binom{5}{1} \times (3x)^4 \times (-3)^1 + \binom{5}{2} \times (3x)^3 \times (-3)^2 \\
&+ \binom{5}{3} \times (3x)^2 \times (-3)^3 + \binom{5}{4} \times (3x)^1 \times (-3)^4 \\
&+ \binom{5}{5} \times (3x)^0 \times (-3)^5 \\
&= 243x^5 - 1215x^4 + 2430x^3 - 2430x^2 + 1215x - 243
\end{aligned}
$$

6 e)

$$
\begin{aligned}
(1+\frac{1}{2}x)^4 = &\binom{4}{0} \times 1^4 \times (\frac{1}{2}x)^0 \\
&+ \binom{4}{1} \times 1^3 \times (\frac{1}{2}x)^1 + \binom{4}{2} \times 1^2 \times (\frac{1}{2}x)^2 \\
&+ \binom{4}{3} \times 1^1 \times (\frac{1}{2}x)^3 + \binom{4}{4} \times 1^0 \times (\frac{1}{2}x)^4 \\
&= \frac{1}{16}x^4 + \frac{1}{2}x^3 + \frac{3}{2}x^2 + 2x + 1
\end{aligned}
$$

6 f)

$$
\begin{aligned}
(2-\frac{x}{4})^5 = &\binom{5}{0} \times 2^5 \times (-\frac{x}{4})^0 \\
&+ \binom{5}{1} \times 2^4 \times (-\frac{x}{4})^1 + \binom{5}{2} \times 2^3 \times (-\frac{x}{4})^2 \\
&+ \binom{5}{3} \times 2^2 \times (-\frac{x}{4})^3 + \binom{5}{4} \times 2^1 \times (-\frac{x}{4})^4 \\
&+ \binom{5}{5} \times 2^0 \times (-\frac{x}{4})^5 \\
&= -\frac{1}{1024}x^5 + \frac{5}{128}x^4 - \frac{5}{8}x^3 + 5x^2 - 20x + 32
\end{aligned}
$$

6 g)

$$\begin{aligned}(2x-\tfrac{1}{x})^4 &= \binom{4}{0}(2x)^4(-\tfrac{1}{x})^0 + \binom{4}{1}(2x)^3(-\tfrac{1}{x})^1 \\ &\quad + \binom{4}{2}(2x)^2(-\tfrac{1}{x})^2 + \binom{4}{3}(2x)^1(-\tfrac{1}{x})^3 + \binom{4}{4}(2x)^0(-\tfrac{1}{x})^4 \\ &= 16x^4 - 32x^2 + 24 - 8x^{-2} + x^{-4}\end{aligned}$$

7)

$$\begin{aligned}(3-2x)^4 &= \binom{4}{0}3^4(-2x)^0 + \binom{4}{1}3^3(-2x)^1 + \binom{4}{2}3^2(-2x)^2 \\ &\quad + \binom{4}{3}3^1(-2x)^3 + \binom{4}{4}3^0(-2x)^4 \\ &= 81 - 216x + 216x^2 - 96x^3 + 16x^4\end{aligned}$$

So $-5(3-2x)^4 = -405 + 1080x - 1080x^2 + 480x^3 - 80x^4$.

8)

$$\begin{aligned}(2-x)^4 &= \binom{4}{0}2^4(-x)^0 + \binom{4}{1}2^3(-x)^1 + \binom{4}{2}2^2(-x)^2 \\ &\quad + \binom{4}{3}2^1(-x)^3 + \binom{4}{4}2^0(-x)^4 \\ &= 16 - 32x + 24x^2 - 8x^3 + x^4 \\ \text{So } (1+2x)(2-x)^4 &= (1+2x)(16 - 32x + 24x^2 - 8x^3 + x^4) \\ &= 16 - 40x^2 + 40x^3 - 15x^4 + 2x^5\end{aligned}$$

9 a)

$$\begin{aligned}(1+\sqrt{3})^4 &= \binom{4}{0}1^4(\sqrt{3})^0 + \binom{4}{1}1^3(\sqrt{3})^1 + \binom{4}{2}1^2(\sqrt{3})^2 \\ &\quad + \binom{4}{3}1^1(\sqrt{3})^3 + \binom{4}{4}1^0(\sqrt{3})^4 \\ &= 1 + 4\sqrt{3} + 6 \times 3 + 4 \times 3\sqrt{3} + 9 = 28 + 16\sqrt{3}\end{aligned}$$

9 b)

$$(\sqrt{2}+\sqrt{5})^4 = \binom{4}{0}(\sqrt{2})^4(\sqrt{5})^0 + \binom{4}{1}(\sqrt{2})^3(\sqrt{5})^1$$

$$
\begin{aligned}
&+ \binom{4}{2}(\sqrt{2})^2(\sqrt{5})^2 + \binom{4}{3}(\sqrt{2})^1(\sqrt{5})^3 + \binom{4}{4}(\sqrt{2})^0(\sqrt{5})^4 \\
&= 4 + 8\sqrt{2}\sqrt{5} + 6 \times 2 \times 5 + 4\sqrt{2} \times 5\sqrt{5} + 25 \\
&= 89 + 28\sqrt{10}
\end{aligned}
$$

9 c)

$$
\begin{aligned}
(\sqrt{7} - \sqrt{2})^5 = &\binom{5}{0}(\sqrt{7})^5(-\sqrt{2})^0 + \binom{5}{1}(\sqrt{7})^4(-\sqrt{2})^1 \\
&+ \binom{5}{2}(\sqrt{7})^3(-\sqrt{2})^2 + \binom{5}{3}(\sqrt{7})^2(-\sqrt{2})^3 \\
&+ \binom{5}{4}(\sqrt{7})^1(-\sqrt{2})^4 + \binom{5}{5}(\sqrt{7})^0(-\sqrt{2})^5 \\
&= 49\sqrt{7} + 5 \times 49 \times -\sqrt{2} + 10 \times 7\sqrt{7} \times 2 \\
&\quad + 10 \times 7 \times -2\sqrt{2} + 5 \times \sqrt{7} \times 4 - 4\sqrt{2} \\
&= 209\sqrt{7} - 389\sqrt{2}
\end{aligned}
$$

10 a)

$\binom{4}{2} \times x^2 \times 5^2 = 6 \times 25 \times x^2$ so 150.

10 b)

$\binom{5}{3} \times (2x)^2 \times (-1)^3 = 10 \times -1 \times 4x^2$ so -40.

10 c)

$\binom{5}{2} \times 8^3 \times x^2 = 10 \times 512 \times x^2$ so 5120.

10 d)

$\binom{6}{2} \times 1^4 \times (-3x)^2 = 15 \times 1 \times 9x^2$ so 135.

10 e)

$\binom{78}{2} \times 1^{76} \times (2x)^2 = 3003 \times 1 \times 4x^2$ so 12012.

11 a)

$\binom{4}{1} \times x^3 \times (-\frac{1}{x}) = 4 \times x^3 \times -\frac{1}{x}$ so -4.

11 b)

$\binom{6}{2} \times (3x)^4 \times (-\frac{2}{x})^2 = 15 \times 81x^4 \times \frac{4}{x^2}$ so 4860.

11 c)

$\binom{18}{8} \times x^{10} \times (-\frac{1}{x})^8 = 43758x^{10} \times \frac{1}{x^8}$ so 43758.

12)

$$\begin{aligned}\frac{(n-1)!}{4!(n-1-4)!}+\frac{(n-1)!}{5!(n-1-5)!} &= \frac{5(n-1)!}{5\times 4!(n-5)!}+\frac{(n-5)(n-1)!}{5!(n-5)!}\\ &= \frac{5(n-1)!+(n-5)(n-1)!}{5!(n-5)!}\\ &= \frac{(n-1)!(5+n-5)}{5!(n-5)!}=\frac{n(n-1)!}{5!(n-5)!}\\ &= \frac{n!}{5!(n-5)!} \text{ as required}\end{aligned}$$

13)

$$\begin{aligned}\frac{(n-1)!}{(r-1)!(n-1-(r-1))!}+\frac{(n-1)!}{r!(n-1-r)!} &= \frac{r(n-1)!}{r\times(r-1)!(n-r)!}+\frac{(n-r)(n-1)!}{r!(n-r)!}\\ &= \frac{r(n-1)!+(n-r)(n-1)!}{r!(n-r)!}\\ &= \frac{(n-1)!(r+n-r)}{r!(n-r)!}=\frac{n(n-1)!}{r!(n-r)!}\\ &= \frac{n!}{r!(n-r)!} \text{ as required}\end{aligned}$$

14)

Each term has $(-3x)^k$ as a factor.

When k is odd, there will be an odd number of negative terms multiplying together, so the result will be negative.

15)

$$\begin{aligned}\binom{n}{2}\times 1^{n-2}\times x^2 &= \frac{n!}{2!(n-2)!}x^2\\ \binom{n}{5}\times 1^{n-5}\times x^5 &= \frac{n!}{5!(n-5)!}x^5\end{aligned}$$

So $\frac{n!}{2!(n-2)!}=\frac{n!}{5!(n-5)!}$ gives

$$\begin{aligned}5!(n-5)! &= 2!(n-2)!\\ 5\times 4\times 3 &= (n-2)(n-3)(n-4)\\ n^3-9n^2+26n-84 &= 0 \text{ So } n=7\end{aligned}$$

16)

$$\binom{n}{4}\times 1^{n-4}\times x^4=\frac{n!}{4!(n-4)!}x^4$$

$$\binom{n}{2} \times 1^{n-2} \times x^2 = \frac{n!}{2!(n-2)!}x^2$$

So $\frac{n!}{4!(n-4)!} = 13\frac{n!}{2!(n-2)!}$ gives

$$\begin{aligned} 2!(n-2)! &= 13 \times 4!(n-4)! \\ 2(n-2)(n-3) &= 312 \\ 2n^2 - 10n - 300 &= 0 \text{ So } n = 15 \end{aligned}$$

17)

$$\binom{n}{2} \times 1^{n-2} \times (2x)^2 = \frac{4n!}{2!(n-2)!}x^2$$
$$\binom{n}{1} \times 1^{n-1} \times (2x)^1 = \frac{2n!}{1!(n-1)!}x$$

So $\frac{4n!}{2!(n-2)!} = 10\frac{2n!}{1!(n-1)!}$ gives

$$\begin{aligned} 4(n-1)! &= 20 \times 2!(n-2)! \\ 4(n-1) &= 40 \text{ So } n = 11 \end{aligned}$$

18)

$$\binom{n}{4} \times 1^{n-4} \times (\tfrac{x}{2})^4 = \frac{n!}{4!(n-4)!} \times \frac{1}{16}x^4$$
$$\binom{n}{5} \times 1^{n-5} \times (\tfrac{x}{2})^5 = \frac{n!}{5!(n-5)!} \times \frac{1}{32}x^5$$

So $\frac{n!}{16\times 4!(n-4)!} = 2\frac{n!}{32\times 5!(n-5)!}$ gives

$$\begin{aligned} 80(n-5)! &= 16(n-4)(n-5)! \\ 5 &= n-4 \text{ So } n = 9 \end{aligned}$$

19 a)

$x^2 \times \binom{12}{6} \times x^6 \times (-\frac{1}{x})^6 = 924x^2$ so 924.

19 b)

$$\binom{10}{2} \times 1^8 \times (4x)^2 - x \times \binom{10}{1} \times 1^9 \times (4x)^1 = 720x^2 - 40x^2 \text{ so } 680$$

19 c)

$$\binom{9}{2} \times 3^7 \times (-x)^2 - 2x \times \binom{9}{1} \times 3^8 \times (-x)^1 = 78732x^2 + 118098x^2 \text{ so } 196830$$

20 a)

$$\begin{aligned}(3+x)^5 &= \binom{5}{0}3^5x^0 + \binom{5}{1}3^4x^1 + \binom{5}{2}3^3x^2 \\ &\quad + \binom{5}{3}3^2x^3 + \binom{5}{4}3^1x^4 + \binom{5}{5}3^0x^5 \\ &= 243 + 405x + 270x^2 + 90x^3 + 15x^4 + x^5\end{aligned}$$

20 b)

$$\begin{aligned}(3-\sqrt{2})^5 &= 243 + 405(-\sqrt{2}) + 270(-\sqrt{2})^2 + 90(-\sqrt{2})^3 \\ &\quad + 15(-\sqrt{2})^4 + (-\sqrt{2})^5 \\ &= 243 - 405\sqrt{2} + 540 - 180\sqrt{2} + 60 - 4\sqrt{2} \\ &= 843 - 589\sqrt{2}\end{aligned}$$

21 a)

$$\begin{aligned}(2+5x)^4 &= \binom{4}{0}(2)^4(5x)^0 + \binom{4}{1}(2)^3(5x)^1 + \binom{4}{2}(2)^2(5x)^2 \\ &\quad + \binom{4}{3}(2)^1(5x)^3 + \binom{4}{4}(2)^0(5x)^4 \\ &= 16 + 160x + 600x^2 + 1000x^3 + 625x^4\end{aligned}$$

21 b)

$(2-5x)^4 = 16 - 160x + 600x^2 - 1000x^3 + 625x^4$.

21 c)

$$\begin{aligned}(2+5x)^4 + (2-5x)^4 &= 32 + 1200x^2 + 1250x^4 \\ \sqrt{75} &= 5x \therefore x = \sqrt{3} \\ (2+\sqrt{3})^4 + (2-\sqrt{3})^4 &= 32 + 1200 \times 3 + 1250 \times 9 = 14882\end{aligned}$$

22 a)

$$\begin{aligned}(1+ax)^8 &= \binom{8}{0}(2)^8(ax)^0 + \binom{8}{1}(2)^7(ax)^1 + \binom{8}{2}(2)^6(ax)^2 \\ &\quad + \binom{8}{3}(2)^5(ax)^3 + \cdots \\ &= 1 + 8ax + 28a^2x^2 + 56a^3x^3 + \cdots\end{aligned}$$

22 b)

$56a^3 = 3 \times 28a^2 \Rightarrow 56a = 84$ so $a = \frac{3}{2}$.

23)

$\binom{6}{2} \times (a)^4 \times (2x)^2 = 60a^4x^2$ so $60a^4 = 4860$ so $a = \pm 3$.

24)

$$\begin{aligned}\binom{n}{1} \times (1)^{n-1} \times (ax)^1 &= nax \\ \binom{n}{2} \times (1)^{n-2} \times (ax)^2 &= \frac{n!}{2!(n-2)!}a^2x^2 \\ &= \frac{1}{2}n(n-1)a^2x^2\end{aligned}$$

So $na = 40$ and $\frac{1}{2}n(n-1)a^2 = 700$.

$$\begin{aligned}\frac{1}{2}n^2a^2 - \frac{1}{2}na^2 &= 700 \\ \frac{1}{2} \times 40^2 - \frac{1}{2} \times 40a &= 700 \text{ so } a = 5 \text{ and } n = 8\end{aligned}$$

25)

$\binom{6}{1} \times (2x)^5 \times (-\frac{1}{ax})^1 = -192(\frac{1}{a^2})x^2$.

$-192(\frac{1}{a}) = -12$.

So $a = 16$.

26 a)

$$\begin{aligned}(2+x)^5 &= \binom{5}{0}2^5(-7x)^0 + \binom{5}{1}2^4(-7x)^1 + \binom{5}{2}2^3(-7x)^2 \\ &\quad + \binom{5}{3}2^2(-7x)^3 + \binom{5}{4}2^1(-7x)^4 + \binom{5}{5}2^0(-7x)^5 \\ &= 32 - 560x + 3920x^2 - 13720x^3 + 24010x^4 - 16807x^5\end{aligned}$$

26 b)

$-560x + 32ax \Rightarrow -560 + 32a = -464$ so $a = 3$.

27 a)

$$\begin{aligned}(1-2x)^9 &= \binom{9}{0}1^9(-2x)^0 + \binom{9}{1}1^8(-2x)^1 + \binom{9}{2}1^7(-2x)^2 + \cdots \\ &= 1 - 18x + 144x^2 + \cdots\end{aligned}$$

27 b)

$1 - 2x = 0.98$ so $x = 0.01$.

$0.98^9 \approx 1 - 18(0.01) + 144(0.01)^2 + \cdots = 0.8344$ so $0.98^9 \approx 0.834$ (3 decimal places).

28)

Set $3 + \frac{2x}{5} = 2.96$ so that $x = -0.1$.

Fully expanding the expression and using the value of $x = -0.1$ in the result gives the value of 2.96^5.

29 a)

$$\begin{aligned}(1-\tfrac{x}{2})^5 &= \binom{5}{0}(1)^5(-\tfrac{x}{2})^0 + \binom{5}{1}(1)^4(-\tfrac{x}{2})^1 + \binom{5}{2}(1)^3(-\tfrac{x}{2})^2 + \cdots \\ &= 1 - \frac{5}{2}x + \frac{5}{2}x^2 + \cdots \\ (2-\tfrac{3x}{4})^6 &= \binom{6}{0}(2)^6(-\tfrac{3x}{4})^0 + \binom{6}{1}(2)^5(-\tfrac{3x}{4})^1 + \binom{6}{2}(2)^4(-\tfrac{3x}{4})^2 + \cdots \\ &= 64 - 144x + 135x^2 + \cdots\end{aligned}$$

29 b)

$1 - \frac{x}{2} = 0.995$, $2 - \frac{3x}{4} = 1.9925$ so $x = 0.01$.

$$\begin{aligned}0.995^5 + 1.9925^6 &= 1 - \frac{5}{2}(0.01) + \frac{5}{2}(0.01)^2 + 64 - 144(0.01) + 135(0.01)^2 + \cdots \\ &= 63.54875 \text{ so } 63.549\end{aligned}$$

30)

$\binom{12}{r} \times (x^2)^{12-r} \times (\frac{1}{x^4})^r$ gives $x^{24} \times x^{-2r} \times x^{-4r} = x^{24}x^{-6r} = x^{24-6r}$.

$24 - 6r = 6(4 - r)$ so powers are always multiples of 6.

Or

$\binom{12}{r} \times (x^2)^r \times (\frac{1}{x^4})^{12-r}$ gives $x^{2r} \times (x^{-4})^{12-r} = x^{2r} \times x^{-48+4r} = x^{6r-48}$.

$64 - 48 = 6(r - 8)$ so powers are always multiples of 6.

31 a)

$$\begin{aligned}(1+3x)^4 &= \binom{4}{0}(1)^4(3x)^0 + \binom{4}{1}(1)^3(3x)^1 + \binom{4}{2}(1)^2(3x)^2 \\ &\quad + \binom{4}{3}(1)^1(3x)^3 + \binom{4}{4}(2x)^0(3x)^4 \\ &= 1 + 12x + 54x^2 + 108x^3 + 81x^4\end{aligned}$$

31 b)

$(1-3x)^4 = 1 - 12x + 54x^2 - 108x^3 + 81x^4$.

31 c)

Hence $f(x) = 2 + 108x^2 + 162x^4$ which has a minimum value of 2.

32 a)

$$\begin{aligned}(5-2x)^6 &= \binom{6}{0}(5)^6(-2x)^0 + \binom{6}{1}(5)^5(-2x)^1 + \binom{6}{2}(5)^4(-2x)^2 + \cdots \\ &= 15625 - 37500x + 37500x^2 - \cdots\end{aligned}$$

32 b)

$(5 - 2x - 6x^2)^6 = (5 - 2(x + 3x^2))^6$.

So $15625 - 37500(x + 3x^2) + 37500(x + 3x^2)^2 - \cdots$

$= -37500 \times 3 + 37500 = -75000$.

32 c)

$$\begin{aligned}(5+3x-2x^2)^6 &= ((5-2x)(1+x))^6 \\ &= (5-2x)^6(1+x)^6 \\ &= (15625 - 37500x + 37500x^2 - \cdots)(1+x)^6 \\ &= (15625 - 37500x + 37500x^2 - \cdots)(1 + 6x + 15x^2 + \cdots)\end{aligned}$$

So the coefficient of the x^2 term is:

$15625 \times 15 - 37500 \times 6 + 37500 \times 1 = 46\,875$.

Alternatively:

$$(5+3x-2x^2)^6 = \left(5 - 2\left(-\frac{3}{2}x + x^2\right)\right)^6$$

$\left(5 - 2\left(-\frac{3}{2}x + x^2\right)\right)^6$.

From part a)

$$= 15625 - 37500\left(-\frac{3}{2}x + x^2\right) + 37500\left(-\frac{3}{2}x + x^2\right)^2 - \cdots$$

$$
\begin{aligned}
&= 15625 + 56250x - 37500x^2 + 37500\left(\frac{9}{4}x^2 - \frac{3}{2}x^3 + x^4\right) - \cdots \\
&= 15625 + 56250x - 37500x^2 + 84375x^2 - 112500x^3 + 37500x^4 \cdots \\
&= 15625 + 56250x + 46875x^2 - \cdots
\end{aligned}
$$

So the coefficient of the x^2 term is:

$15625 \times 15 - 37500 \times 6 + 37500 \times 1 = 46\,875.$